NETWORKING

Software-Defined Networking

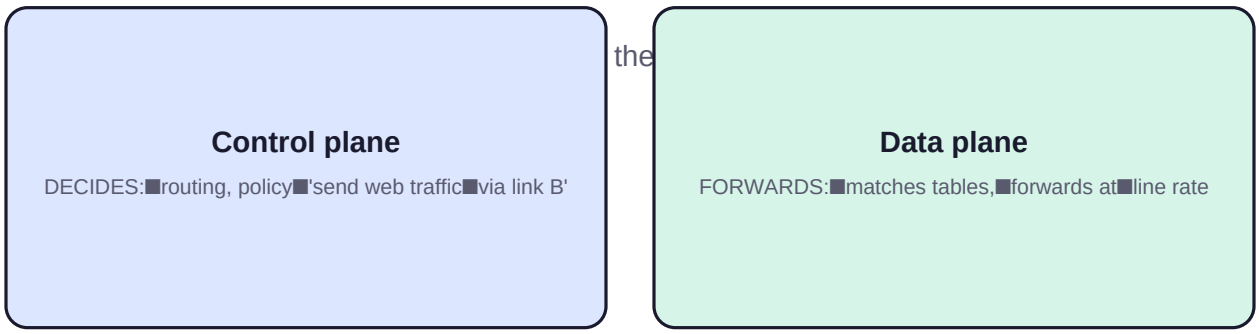
Control plane, meet software — networks programmed like code.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/5

Control Plane vs Data Plane

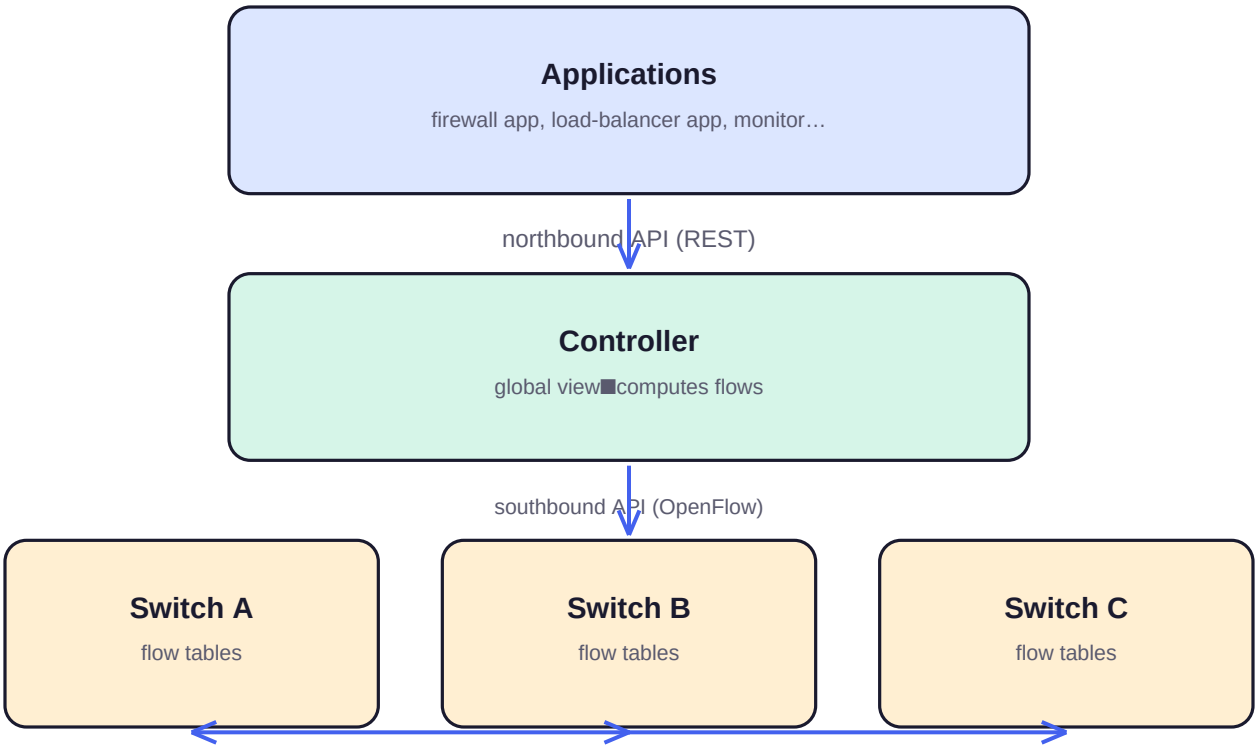


- Traditional routers bundle both: every box runs OSPF/BGP and forwards — distributed, autonomous, hard to coordinate.
- SDN centralizes DECISIONS in a controller; switches become fast, dumb forwarding engines.
- Result: network-wide policy from one program instead of box-by-box CLI.

STEP 2/5

The SDN Architecture

Three layers, two APIs.



Apps express intent northbound; the controller programs flow tables southbound.

STEP 3/5

OpenFlow: Programming the Switch

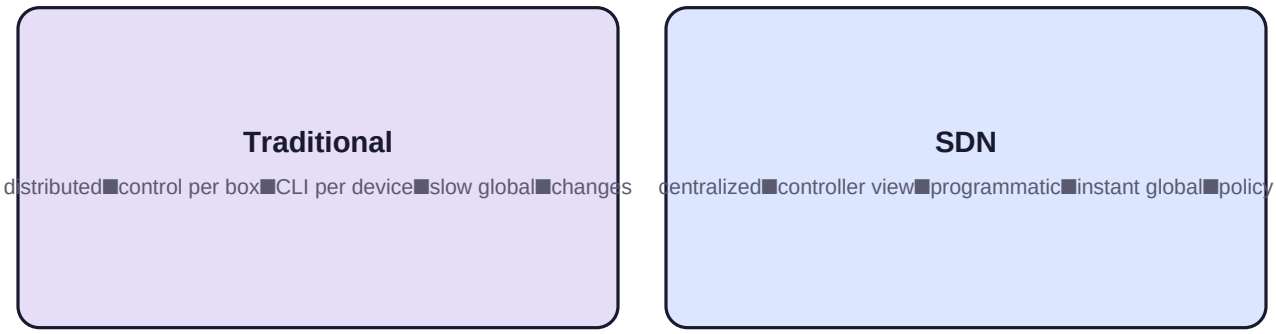
Flow tables are match → action rules installed by the controller.

```
match: { in_port: 1, eth_type: IPv4, tcp_dst: 80 }
action: forward → port 3        // a 'flow'
match: { in_port: 1, * }
action: send → controller       // unknown: ask the brain
```

- First packet of a new flow goes to the controller ('packet-in'); the rule it installs handles the rest.
- Flows have timeouts and counters — the controller sees traffic stats per rule.
- Reactive (learn as traffic arrives) vs proactive (pre-install known policies).

STEP 4/5

Traditional vs SDN



- SDN strength: consistent policy, automation, global optimization (traffic engineering).
- SDN risk: the controller is a single point of failure AND attack — cluster it, secure it.
- Reality check: most 'SDN' in production is overlay-based (VXLAN) or intent APIs, not pure OpenFlow.

STEP 5/5

Where SDN Actually Won

The ideas succeeded even where OpenFlow didn't.

Datacenter fabrics

Clos topologies with centralized routing — every hyperscaler runs this.

SD-WAN

Policy-based path selection across internet/MPLS — branch offices without telco lock-in.

Cloud overlays

VPCs are SDN: virtual networks programmed over shared physical fabric (VXLAN).

5G / telco

Network slicing and programmable forwarding in carrier cores.

STEP ✓/✓

Key Takeaways

SDN — Remember

- SDN separates the control plane (decisions) from the data plane (forwarding).
- Controller has the global view; switches hold flow tables programmed via OpenFlow.
- Apps express intent northbound; flows are match→action rules with counters.
- Wins: consistent policy, automation, global optimization — at the cost of controller risk.
- SDN's real victories: datacenter fabrics, SD-WAN, and cloud VPC overlays.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev